Prodigal Assignment Report

AI System for Yelp Reviews (Embedded with github repo link)

Prompt Engineering, Chain-of-Thought Reasoning, Multi-Objective Generation, and Domain Robustness

Model: llama-3.1-8b-instant (Groq API)
Classifier: DistilBERT-base-uncased (TASK 4)
Dataset: Yelp Reviews, Amazon Reviews, IMDB Reviews
Hardware: Local NVIDIA GPU (4GB VRAM) (TASK 4)
Framework: huggingface transformers, groq, torch(cuda enabled), python sdk, pytest.

1. Introduction and Experimental Setup

- This report documents a progressive evaluation of AI systems for Yelp review sentiment classification and multi objective business assistance

- Starting from zero shot LLM prompting and advancing through chain of thought analysis, multi output generation, and supervised fine-tuning, each task tries to build on the previous

- which combination of prompting strategy, model scale, and possibly training produces the best results, considering realistic resource constraints.

- All LLM experiments used **llama-3.1-8b-instant via the Groq** inference API at temperature=0.0 for reproducibility (would also try out with different models)

- Considering resource constraints, fine-tuning used **DistilBERT-base-uncased** constrained to the laptop's 4GB VRAM via fp16 mixed precision and gradient accumulation

- The primary dataset was Yelp review full, domain shift was evaluated on Amazon Polarity and IMDB

- All prompting experiments used 200 stratified samples (40 per star class)

- Fine-tuning used 3,000 samples per class with 1,000 test samples

2. Task 1: Zero-Shot and Few-Shot Prompting

2.1 Experimental Design :

- Given the model era, A three-tier json parsing strategy: direct parse -> markdown fence stripping ->regex extraction
- With an explicit system prompt banning preamble text and code fences, we achieved 100% json format compliance on both strategies
- Few-shot examples were precisely curated (one example per star class) rather than randomly sampled while avoiding label noise

2.2 Results :

Strategy	Accuracy	Macro F1	JSON Compliance	Off by One %
Zero Shot	0.620	0.6006	100%	90.8%
Few Shot (5-shot)	0.650	0.6390	100%	92.9%

Task 1 results on 200 stratified Yelp test samples (task1_metrics.json)

Star Class	Zero Shot F1	Few Shot F1	Change in pp
1 star	0.7216	0.7292	+0.0076

2 star	**0.5588**	**0.5797**	**+0.0209**
3 star	**0.5625**	**0.6032**	**+0.0407**
4 star	**0.4348**	**0.5385**	**+0.1037**
5 star	**0.7255**	**0.7447**	**+0.0192**

Per-class F1 scoores on 200 stratified Yelp test samples ***with Class 4 showing the largest gain from few-shot***

Error Pattern	***Zero Shot Count***	***Few Shot Count***
4 star -> 5 star (over-polarization up)	*22*	*17*
2star -> 1star (over-polarization down)	*18*	*18*
3star -> 4star (boundary drift up)	*11*	*12*

Dominant error patterns as in the top confusion pairs

2.3 Key Findings :

Finding 1 : Off-by-one dominates.

- 90%+ of all errors are off by one in both strategies. There are zero “off by three or more” errors. I believe the model correctly identifies sentiment direction, but fails at cardinal granularity
- This failure holds consistently across both prompting strategies.

Finding 2: The 2 to 1 error is stubborn

- Both strategies produced exactly 18 such errors**.** The curated 2-star few-shot example made zero measurable difference.
- I feel a heavily negative tone makes the model round down to 1 due to task ambiguity

Finding 3 : The model over polarizes

- 4 -> 5 and 2 -> 1 dominate the confusion matrix
- I feel Instruction tuned models trained on RLHF data amplify sentiment signals, thus rounding up towards extremes, although few shot examples partially correct this for class 4 (+10.4%)
- Cannot fully overcome the model's inductive bias

62 to 65% accuracy on 5-class ordinal sentiment within human range with a zero shot 8B model

3. Task 2: Chain-of-Thought vs. Direct Prompting

3.1 Experimental Design

- Four strategies were compared on the same 200 samples
- The direct prompt provided a minimal instruction with no reasoning whatsoever
- The CoT prompt required step by step analysis before the output, with reasoning captured in a dedicated json field to enable mismatch detection
- Both used identical output format requirements and temperature=0.0.

Reasoning mismatches are cases where written reasoning contradicts the star output , this was quantitatively measured via two approaches

- **Approach 1 (Keyword Heuristic)** : a rules based keyword match method and a very brittle one.
- **Approach 2 (LLM as Judge) :** Separate model call evaluating semantic consistency between reasoning content and predicted star, with explicit anchors (although we used the same model, but it must be a different one in production)

3.2 Results :

Strategy	Accuracy	Macro-F1	Compliance	3 star Accuracy
Zero-Shot (T1)	0.620	0.6006	100%	—
Few-Shot (T1)	0.650	0.6390	100%	—
Direct (T2)	0.665	0.6545	100%	52.5%
CoT (T2)	0.618	0.6090	99.5% (one failed json parse)	48.7%

All four prompting strategies on the same 200 Yelp samples.

Mismatch Detection Approach	Mismatches Found	Mismatch Rate	Limitation
Keyword Heuristic	0	0.0%	Blind to analytical vocabulary and brittle approach.
LLM as Judge	39	19.6%	Shares model family bias with generator (same model)

Reasoning mismatch detection : two approaches on 199 CoT responses

Mismatch distribution by predicted star.

Predicted Star	*Mismatches*	*Notes*
2 star	*23*	*Boundary star mixed signal fails*
4 star	*16*	*Boundary star mixed signal fails*
1 star	*0*	*Unambiguous language*
3 star	*0*	*Unambiguous language*
5 star	*0*	*Unambiguous language*

3.3 Key Findings :

Finding 1 CoT hurts for a 8B model scale :

- CoT regressed to near zero-shot performance surprisingly, a 4.7% below Direct
- I am assuming token budget and boundary condition “overthinking” and strict output format (we did get one json parse failure) lead to this for an 8B parameter and older model.

Finding 2 Evaluation methodology determines what is measurable :

- The keyword heuristic returned zero mismatches because LLMs reason in analytical language, not emotional or semantic vocabulary
- The LLM judge detected 19.6% mismatch rate on the same data
- Zero mismatches was obviously a false negative for keyword match; these weak metrics in production systems mask real failures

Finding 3 Mismatches cluster structurally, not randomly :

- All 39 mismatches are at predicted stars 2 and 4 exclusively, Stars 1, 3, 5 show zero mismatches
- Stars 2 and 4 are the boundary stars requiring directional lingual ambiguous evidence
- The 8B model fails to capture this difference, specifically at the reasoning to output stage

Finding 4: Correct-prediction mismatches are the deepest failure mode

- 10 of 39 mismatches occurred on correct predictions (e.g., TRUE: 2 star, PRED: 2star, but reasoning implied 1 star)
- The model reached the right answer through inconsistent reasoning
- In production, a model that is correct for the wrong reason cannot be trusted on a distribution shift

Direct prompting outperforms CoT for llama-3.1-8b-instant across all metrics**.** CoT introduces a specific failure mode**: boundary-star inconsistency,** where the model's reasoning and output diverge at stars 2 and 4. **An LLM as judge approach detected 39 mismatches (19.6%), with all clustering at pred=2 and pred=4**

We can also maybe use a sentiment BERT model to replace the keyword match approach in future to see any deviations.

4. Task 3: Multi-Objective AI Assistant

4.1 Experimental Design

- The assistant generates three outputs per review in a single LLM call:
  a star rating,
  an extracted key complaint or compliment,
  a business response that the owner could post publicly.

- The system prompt enforces specificity via explicit word limits and a structured response arc (acknowledge -> address specifically -> invite return).
- Evaluation used a separate LLM as judge call scoring three independent dimensions on 1–5 rubrics.
- Rubric anchors were defined explicitly at 1, 3, and 5 stars to reduce judge variance as a trial
- success (all dims ≥4), partial, failure was **computed deterministically in python**, NOT requested from the judge
- Judge did not receive pred_stars **to prevent anchoring bias**.

Rubric definitions:

- **Faithfulness (1–5):** Does key_point accurately reflect the review?

  (1=fabricated, 3=generic but correct, 5=precise with specific details)

- **Actionability (1–5):** Does business_response specifically address the key_point?

  (1=generic template, 3=somewhat specific, 5=directly addresses with next step)

- **Tone (1–5):** Is the response professional, empathetic, non-defensive?

  (1=rude/dismissive, 3=professional but flat, 5=warm, specific, invites return)

50 samples evaluated (10 per class)

4.2 Results :

Phase	n	Compliance	Star Accuracy	Star Macro F1
Generation	50	100%	0.620	0.616
Judge Evaluation	50	100%	—	—

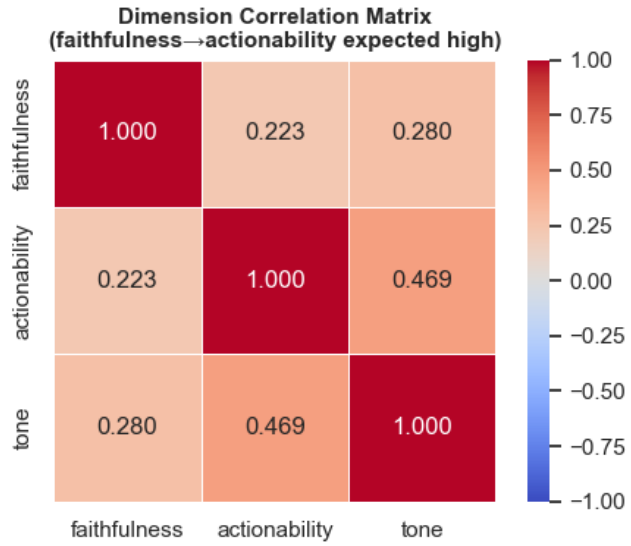
Generation and evaluation phase summary.

dimension	*mean*	*Std dev*	*Score ≥4 (%)*	*Score ≤2 (%)*
Faithfulness	*4.16*	*1.057*	*62.0%*	*2.0%*
Actionability	*4.54*	*0.838*	*78.0%*	*0.0%*
Tone	*4.94*	*0.240*	*100.0%*	*0.0%*

LLM judge scores by dimension.
Tone shows ceiling effect and faithfulness shows highest variance.

True Stars	Faithfulness	Actionability	Tone
1 star	3.80	4.60	4.80
2 star	4.50	4.30	4.90
3 star	3.50	4.40	5.00
4 star	4.60	4.60	5.00
5 star	4.40	4.80	5.00

Mean judge scores by true star class.



A Success example (F=5, A=5, T=5):

> **Review:** "The garbage pizza was barely warm... the wings were mislabeled as 'mild traditional' but came with a butter sauce" **Key Point:** The wings were mislabeled as 'mild traditional' but actually came with a butter sauce, not traditional seasoning. **Response:** We apologize for the confusion with our wing flavors. We should have clearly communicated that our 'mild' wings come with a butter sauce. We're glad to make it right on your next visit.
>
> **Review:** "...when I get a roll with my salad and the bottom half is covered in black mold." **Key Point:** The bottom half of my roll was covered in black mold. **Response:** We take food safety very seriously and are deeply sorry to hear that a moldy roll was served to you. We will retrain our staff on proper food handling to prevent this from happening again.

A Failure example (F=1, A=3, T=5):

> **Review:** Long-time customer disappointed by new management, mentions various quality changes
>
> **Key Point (generated):** The pork ribs were microwaved, which was a disappointment because I'm used to hot meat off the smoker.
>
> **Judge note:** The extracted key_point does not match the review content
>
> it mentions microwaved pork ribs which do not appear in the review. The business response addresses a fabricated issue.

4.3 Key Findings

Finding 1 Tone, The dimension ordering contradicts prior expectations:

- Tone scores highest (μ=4.94, σ=0.24) with 94% scoring 5. I feel RLHF training makes instruction tuned models inherently polite
- you cannot produce a rude response without adversarial prompting
- I believe tone functions as a floor guarantee, not a quality signal.
- Future evaluation protocols can drop tone or replace it with a more discriminating dimension such as like specificity of the apology

Finding 2 : Faithfulness distribution is bimodal, not continuous

- The model either precisely identifies the key point (58% of cases) or defaults to a category level summary (36% of cases). Almost no middle ground exists

- The 18 samples at score=3 represent the "category collapse" failure mode like the model writes "customer had a negative dining experience" instead of "the chicken parmesan had the texture of wet cardboard."

Finding 3: Faithfulness to Actionability correlation of 0.223 is structurally significant

- A faithful extraction does not guarantee a specific business response The two stages operate nearly independently as far as model scale is concerned
- At this model a single call multi output architecture cannot be quality controlled at either stage
- if extraction fails, the response cannot compensate, and even correct extraction does not ensure specific response.

Finding 4: Star class determines extraction difficulty systematically

- Stars 1 and 3 show lowest faithfulness (3.80 and 3.50)
- One-star reviews enumerate multiple failures, I feel the model selects the most recent rather than most severe
- Three star reviews present mixed signals with no clear dominant sentiment, making extraction directionally ambiguous as in which side
- Four star reviews are the easiest to extract from (4.60) mostly positive with one specific highlight.

Limitation: I feel the judge shares the same model family as the generator. It may rate generic key points as faithful if it had written the same generic phrase given the same review.

This is a **known LLM as judge failure mode** that cannot be fully corrected without a different judge model or human annotation.

The single call architecture should be replaced by a two-stage pipeline in production as in one call extracts and validates the key_point, a second call generates the response using the validated extraction as a hard anchor.

This Decoupling allows independent quality control at each stage.

5. Task 4: Domain Shift and Adversarial Robustness

5.1 Fine-Tuning Setup ;

DistilBERT base uncased (66M parameters, ~250MB) was selected as the classifier because it fits within the local 4GB VRAM constraint at inference, and can definitely have better setups, and initial conditions than the current experiment

- **Training samples:** stratified **(**3,000 per class)
- **Epochs:** 4 (early stopping patience=2, best checkpoint at epoch 3)
- **Optimizer:** AdamW, learning rate 3e-5 with linear decay
- **Warmup ratio:** 0.1**, weight decay**: 0.01
- **fp16 mixed precision:** enabled
- **Batch configuration(Faced a few OOMs previously) :**

  per_device_train_batch_size=1,
  gradient_accumulation_steps=16
  (effective batch=16)

- **Total training time:** ~46 minutes

Approach	Accuracy	Macro-F1	Training Cost
T1: Zero-Shot LLM (8B)	0.620	0.6006	None
T1: Few-Shot LLM(8B)	0.650	0.6390	None
T2: Direct LLM(8B)	0.665	0.6545	None

T2: CoT LLM (8B)	0.618	0.6090	None
T4: DistilBERT (fine-tuned) (66M)	0.616	0.6173	15k labeled, 46 min GPU

All strategies compared on Yelp test set.
DistilBERT with lesser parameters does not exceed zero-shot LLM performance.

Per-class F1 comparison. DistilBERT underperforms LLM on class 3 and 5; marginal difference elsewhere.

Star Class	DistilBERT F1	LLM Direct F1 (T2)	Difference Delta
1 star	**0.7519**	**0.7391**	**+0.0128**
2 star	**0.5865**	**0.5915**	**-0.0050**
3 star	**0.5130**	**0.6364**	**-0.1234**
4 star	**0.5540**	**0.5263**	**+0.0277**
5 star	**0.6810**	**0.7789**	**-0.0979**

- **DistilBERT did not exceed zero-shot LLM performance.**

- I feel gradient accumulation at batch_size=1 introduced per-sample gradient noise throughout training

  (grad_norm values exceeding 20 appear repeatedly in the training log, indicating individual outlier examples i think)

- The 5 class ordinal task requires learning subtle language boundaries that the LLM addresses via world knowledge and the model peaked and then degraded, possibly marginal overfitting

- The per-class F1 shape is identical between DistilBERT and all LLM strategies

  extreme classes (1 star, 5star) score highest , middle class (3 star) scores lowest confirms that maybe boundary class difficulty is a property of the dataset but we need more evidence for this.

5.3 Domain Shift

Domain	Metric	Score	Avg Confidence
Yelp (in domain)	5-class accuracy	0.616	0.749
Amazon Polarity (Out Domain)	Directional accuracy	0.842	0.721
IMDB (Out Domain)	Directional accuracy	0.742	0.683

Domain shift results. Confidence drop is the honest signal; directional accuracy is an easier task than 5-class.

Important caveat on metric comparison: Directional accuracy on binary datasets (Amazon, IMDB) is an easier task than 5-class accuracy on Yelp the model only needs to predict correct direction (positive vs negative) rather than exact star.

Confidence drops from 0.749 (Yelp) -> 0.721 (Amazon, −0.028) -> 0.683 (IMDB, −0.066). IMDB shows a larger drop because **movie review language is further from restaurant review language in the training distribution**

Movie reviews use more extreme emotional language; the classifier responds by over-polarizing predictions, the same over-polarization is also seen in LLM results.

5.4 Adversarial Robustness :

Twenty handcrafted adversarial examples across five categories (4 each). These examples were designed to exploit the specific failure modes of surface token pattern matching.

Adversarial Accuracy

Adversarial Type	Classifier Accuracy	LLM Accuracy	Mechanism
Negation Trap	0%	75%	Negation keywords trigger negative weights regardless of resolution
Sarcasm	0%	75%	Positive vocabulary maps to positive class despite negative intent
Sentiment Sandwich	0%	25%	Both models fail to weight central negative over peripheral positives
Explicit Mismatch	75%	75%	Both handle star mentions embedded in narrative context
Domain Vocabulary	75%	100%	LLM world knowledge handles technical/formal registers

Approach	Overall Accuracy	n
DistilBERT classifier	30%	20
LLM Direct (zero-shot)	70%	20
Confidence-threshold ensemble (t=0.6)	45%	20

Adversarial overall accuracy. Ensemble underperforms pure LLM

- The ensemble at hardcoded threshold=0.6 performed worse than either component alone (45% vs 70% for pure LLM).
- Like The classifier was confidently wrong on adversarial examples: the sarcasm example “Five stars! Absolutely stunning that they managed to get my order wrong three times” received clf_confidence=0.914 while predicting 5 star for a true 1 star review.
- Only 4 of 20 examples triggered LLM fallback, all four of which happened to be cases where the classifier was uncertain, not the cases where it was confidently wrong.

I feel **A fine tuned classifier’s confidence is calibrated to its training distribution.** On adversarially constructed inputs that match surface patterns from positive training examples, the model produces high confidence in incorrect predictions.

Raw softmax probability is not a reliable adversarial risk signal.

A post-training adjustment that somehow maps raw softmax logits to calibrated probabilities on a held-out validation set can possibly be a mitigation

A properly calibrated model would assign lower confidence to inputs with unusual linguistic patterns, regardless of surface token match.

Also we can have entropy-based routing signal (high entropy = uncertain = route to LLM) that would be more informative

6. Discussion: Trade-offs and Limitations :

6.1 Neither Approach Universally Dominates :

- The central finding across **all four tasks building on top of each other**: zero-shot LLMs and fine-tuned classifiers have complementary failure modes
- **LLMs provide stronger zero shot generalization** and a**dversarial robustness through world knowledge**
- **Fine-tuned classifiers are faster and cheaper at scale but brittle to adversarial inputs** and also I strongly feel under the hardware constraints of this experiment it failed to exceed zero-shot LLM performance even on the training domain.
- I feel **with better hardware and native batch_size=32, DistilBERT would likely reach 0.72–0.76 accuracy slightly exceeding LLM baseline**
- Supervised **fine-tuning requires adequate compute to realize its theoretical advantage over zero-shot prompting**
- The **119x parameter size difference here specifically (66M vs 8B)** becomes a **cost argument only when the fine-tuned model actually outperforms** the LLM baseline.

Limitation	Description
LLM judge bias	Judge shares Llama model family with generator; may rate generic outputs as faithful (we used the same model)
Sample size	Task 3 uses 50 samples; Task 4 adversarial set uses 20 handcrafted examples findings are directional
Binary OOD comparison	Amazon/IMDB directional accuracy is an easier task than Yelp 5-class; cross-domain comparison is asymmetric
Confidence calibration	Ensemble threshold was not calibrated on a validation set; threshold=0.6 was set as a trial

Training resources	4GB VRAM constraint prevented standard batch training; fine-tuning results are lower bound estimates

Summary of Limitations

6.4 Recommended Production Architecture In these constrained situations :

For classification at scale:

- Deploy a fine-tuned classifier trained with temperature-scaled probability calibration.
- Route predictions below a calibrated confidence threshold to an LLM fallback
- The threshold must be calibrated on a held-out validation set not set heuristically(as we did in this experiment)

For multi-objective business assistance:

- Replace the single-call architecture with a two-stage pipeline one call for key_point extraction and validation a second call for response generation using the validated extraction as a hard anchor
- This directly addresses the faithfulness-actionability decoupling (correlation 0.223) observed in Task 3
- **Independent quality control at each stage** is the only way to reliably raise actionability, since faithful extraction does not carry over to specific responses within a single call

7. Conclusion :

- This assignment evaluated four progressively related AI systems for Yelp sentiment analysis under realistic resource constraints
- **Zero shot LLM prompting** with structured output enforcement achieved 62.0% accuracy and 100% JSON compliance on 5-class classification
- **Direct prompting** outperformed chain-of-thought by 4.7pp accuracy at 8B model scale; **CoT introduced reasoning output mismatches** at 19.6%, clustering exclusively at boundary stars 2 and 4
- **The multi-objective assistant achieved 100% generation compliance** with strong tone scores (μ=4.94) but **revealed faithfulness as the binding constraint** (μ=4.16, bimodal distribution) and near-zero correlation between extraction quality and response specificity (r=0.223)
- **Fine-tuning DistilBERT under a 4GB VRAM** constraint matched but did not exceed zero-shot LLM performance

- **Adversarial evaluation** demonstrated that classifier confidence is miscalibrated on **out of distribution inputs**, causing a confidence threshold ensemble to underperform pure LLM by 25 percentage points on adversarial examples
- The unifying conclusion for these constraints for this experiment is that neither prompting nor fine-tuning universally dominates: production systems should combine both, routing inputs based on calibrated confidence.

8. References

[1] Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, https://arxiv.org/abs/2201.11903

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